



نظم التشغيل (خاص)

11:30-12:30

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Course code: CS321

3rd Level

Midterm Exam

Duration: One hour

This exam for the following program(s) : (Add the program's name(s))

 **Will be reviewed**

8. Which of the following is NOT a state of a process in the five-state model? 

☒ Running



✓ Will be reviewed

9. What is the primary purpose of a Mutex in a multithreaded environment?

To ensure mutual exclusion for accessing shared resources.

To prioritize thread execution based on importance.

To manage memory allocation for threads.

To synchronize the execution of all threads.

✓ Will be reviewed

10. Compared to mutexes, semaphores offer which additional capability?

Faster synchronization

More complex implementation

Ability to control access to a limited number of resources

None of the above

✍ **Will be reviewed**

11. Semaphores are used for: 📖

Memory management

Process scheduling

Deadlock prevention

Process synchronization

✍ **Will be reviewed**

12. Which of the following hardware instructions is commonly used to implement critical sections in a multiprocessor system? 📖

TestAndSet

Load

Move

 Add

 **Will be reviewed**

13. DMA transfers are most beneficial for: 

 Frequent, small data exchanges between CPU and devices.

 Large, bulk data transfers between devices and memory.

 Real-time applications requiring minimal latency.

 All of the above.

 **Will be reviewed**

14. What is a key characteristic of a race condition? 

 The outcome of program execution depends solely on the order of instructions within a single process.

 The operating system runs out of memory during program execution.

 Multiple processes compete for access to a shared resource without proper synchronization.

 An infinite loop occurs within a single process.

📝 **Will be reviewed**

15. What is a short-term scheduler? 🗒️

It selects which process must be brought into the ready queue

It selects which process must be executed next and allocates CPU

It selects which process must be removed from memory by swapping

None of the mentioned

📝 **Will be reviewed**

16. When a program needs to perform a privileged operation like reading a file, it switches from which mode to which mode? 🗒️

Application Mode to System Mode

Guest Mode to Host Mode

User Mode to Kernel Mode

Kernel Mode to User Mode

✓ Will be reviewed

17. What is the primary cause of priority inversion in a multitasking system? 📄

Insufficient memory allocation

Interrupt latency exceeding a specific threshold

A high-priority process waiting on a resource held by a lower-priority process

Deadlock between processes

✓ Will be reviewed

18. What is the disadvantage of using busy waiting in a solution to the critical-section problem? 📄

It requires complex data structures for managing process states.

It wastes CPU cycles while a process waits to enter its critical section.

It guarantees mutual exclusion for accessing shared resources.

It is prone to deadlocks.

✓ Will be reviewed

19. What is the primary purpose of an interrupt? 

 To terminate a running program

 To signal the CPU about an event requiring attention

 To perform a system call

 To allocate memory for a new process

 **Will be reviewed**

20. The primary function of a scheduler is to: 

 Manage the creation and deletion of processes

 Determine which process gets the CPU next

 Manage I/O devices

 Allocate memory to processes

 **Will be reviewed**

21. What happens when threadA tries to acquire a Mutex that is already locked by threadB? 

 The threadB is immediately terminated.

The operating system throws an error and the program crashes.

The threadA is blocked until the threadB releases the Mutex.

The threadA preempts the threadB and acquires the Mutex.

✓ **Will be reviewed**

22. Which mode has full access to system resources like memory and devices? 

Application Mode

User Mode

Kernel Mode

Guest Mode

✓ **Will be reviewed**

23. What is the primary purpose of synchronization mechanisms in an operating system? 

To handle device interrupts and manage I/O operations.

To prevent race conditions and ensure data consistency when multiple processes access shared resources.

To prioritize processes based on their importance.

● To allocate memory efficiently to running processes.

📝 **Will be reviewed**

24. How does DMA differ from programmed I/O? 📖

DMA is slower than programmed I/O.

DMA transfers data in smaller chunks.

DMA utilizes the system bus without CPU intervention.

DMA involves direct interaction between the CPU and device.

📝 **Will be reviewed**

25. The degree of multiprogramming refers to: 📖

The speed at which the CPU can switch between programs.

The number of processes that can reside in memory at the same time.

The number of I/O devices available in the system.

The number of users that can access the system simultaneously.

 **Will be reviewed**

26. Which of the following operations is performed on a semaphore? 

 Read

 Write

 P (wait) and V (signal)

 All of the above

 **Will be reviewed**

27. What happens when a DMA transfer is complete? 

 The operating system takes over control of the data transfer.

 The CPU automatically initiates a new transfer without notification.

 The DMA controller generates an interrupt to inform the CPU.

 The DMA controller shuts down and requires manual restart.



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